

AI Powered Customer Insight Engine for Beauty Brands



Problem Statement

- Beauty brands like Fenty launch diverse product lines but struggle to monitor real-time customer sentiment.
- Reviews are essential but manually analyzing thousands of them is not scalable.
- Key issues, praise points, and sentiment trends are often missed in real-time.

Goal:

Develop an AI-powered system to extract structured insights from unstructured customer feedback.



Proposed AI Solution

An AI driven framework for Customer Insight

- **Sentiment Classification**
 - **What We Did:** Used the RoBERTa model to label 1,000 reviews as Positive, Neutral, or Negative.
 - **Why RoBERTa:** A modern language model built on BERT, trained on more data, and better at understanding tone in short reviews.
- **Review Cleaning & Consolidation**
 - Combine pros and cons fields into a single, cleaned opinion column.
- **Topic Modeling**
 - **What We Did:** Used NMF to uncover 5 main themes in customer feedback.
 - **Why NMF:** Ideal for short, informal texts like product reviews making it easier to interpret.
- **Insight Generation**
 - Merge sentiment with product metadata to detect rating mismatches and flag underperformers



Data Overview

Customer Reviews (cleaned_makeup_reviews.csv)

- 1,000 reviews
- Fields: cleaned text (**truncated_review**), sentiment, product_name, brand

Product Metadata (cleaned_makeup_products.csv)

- Fields: rating, num_shades, product_name, brand

Cleaning & Merging

- Standardized brand and product names (lowercase, no extra spaces)
- Unified dataset enabled comparison of sentiment vs. product features



Methodology

AI Pipeline Workflow

1. Merge Datasets

→ Combine customer reviews and product metadata using **product_link_id**

2. Clean and Prepare Reviews

→ Merge pros & cons into a single opinion column

→ Apply NLP preprocessing: lowercase, remove punctuation, strip spaces

3. Sentiment Classification (RoBERTa)

→ Classify each review as Positive, Neutral, or Negative

4. Topic Modeling (NMF)

→ Use CountVectorizer + NMF to uncover 5 major review themes

→ Example topics: texture, wearability, shade match



Results and Insights: Sentiment Classification & Topic Modeling

Sentiment Classification

- Used cardiffnlp/twitter-roberta-base-sentiment on 1,000 customer reviews.
- Categorized reviews as **positive**, **neutral**, or **negative**.
- Automated classification helped quantify brand perception.
- Saved time and preserved customer expression nuances.

Topic Modeling

- Applied **Non-negative Matrix Factorization (NMF)** to extract key themes:
 - Shade match
 - Hydration
 - Wearability
 - Coverage
 - Packaging
- Revealed deeper insight into what customers care about most.
- Supported product teams in identifying areas for improvement.





Results and Insights- Continued

Model Evaluation

- **Accuracy: 93.8%**
- **F1-Score:**
 - **Positive: 0.95**
 - **Neutral: 0.93**
 - **Negative: 0.00** (*due to only 2 negative reviews*)
- **Note: Class imbalance limited performance on negative sentiment.**
- **Future iterations should focus on balanced datasets.**

Brand Sentiment Trends

- **Positive leaders: IT Cosmetics, Benefit, NUDESTIX**
- **Neutral majority: NARS, L'Oréal**
- **Negative mentions: bareMinerals, BOBBI BROWN**
- **Trends visualized using monthly heatmaps & 3-month rolling averages:**
 - **NARS improved sharply in 2025.**
 - **Lancôme and Laura Mercier showed volatility.**



Takeaway: Brands can tailor engagement based on real-time sentiment shifts.





Bias Detection and Fairness

Some brands or shade ranges may be over- or underrepresented

Review tone can vary by demographic and region

Mitigation Strategies:

- Use more diverse datasets
- Analyze sentiment across shade groups
- Flag consistently low sentiment products for review



Future Implementation and Challenges

Future Implementation

- **Expand dataset for better sentiment balance**
- **Fine-tune RoBERTa on domain-specific reviews**
- **Deploy live Streamlit app with API data flow**
- **Add demographic & product category filters**
- **Integrate LLMs for smarter insight generation**

Challenges

- **Class imbalance: Only 2 negative reviews**
- **Streamlit in Colab: Session timeouts & manual uploads**
- **No demographic data to track sentiment across subgroups**
- **Limited time for dashboard deployment and testing**



Thank you!

